Algebra 1 — Arithmetic & Geometric Sequences — Answer Key

1. Rule A(n) = 4 + (n−1)(−5): 3rd term = −6, 5th = −16, 10th = −41

2. Rule A(n) = 1.4 + (n−1)(3): 3rd term = 7.4, 5th = 13.4, 10th = 28.4

3. (5, 13, 21, 29, …) Recursive: A(1) = 5, A(n) = A(n−1) + 8 · Explicit: A(n) = 5 + 8(n−1)

4. (51, 47, 43, 39, …) Recursive: A(1) = 51, A(n) = A(n−1) − 4 · Explicit: A(n) = 51 − 4(n−1)

5. (0.2, 0.5, 0.8, 1.1, …) Recursive: A(1) = 0.2, A(n) = A(n−1) + 0.3 · Explicit: A(n) = 0.2 + 0.3(n−1)

6. (1.6, 0.8, 0, −0.8, …) Recursive: A(1) = 1.6, A(n) = A(n−1) − 0.8 · Explicit: A(n) = 1.6 − 0.8(n−1)

7. The friend multiplied the difference by 8 instead of (8 − 1) = 7 — the explicit formula uses (n − 1). Correct: A(8) = 3 + (8−1)(5) = 38.

8. Neither — the differences (1, 2, 3) are not constant, and the ratios are not constant either.

9. Geometric — common ratio 1/2.

10. Arithmetic — common difference 3.

11. (5, 15, 45, 135, …) Recursive: A(1) = 5, A(n) = 3·A(n−1) · Explicit: A(n) = $5 \cdot 3^{n-1}$

12. (−2, −12, −72, −432, …) Recursive: A(1) = −2, A(n) = 6·A(n−1) · Explicit: A(n) = $-2 \cdot 6^{n-1}$

13. (500, 100, 20, 4, …) Recursive: A(1) = 500, A(n) = (1/5)·A(n−1) · Explicit: A(n) = $500 \cdot (1/5)^{n-1}$

14. (75, 15, 3, 3/5, …) Recursive: A(1) = 75, A(n) = (1/5)·A(n−1) · Explicit: A(n) = $75 \cdot (1/5)^{n-1}$

15. Photo copy: $8 \times (0.88)^5 \approx 4.22$ inches

16. f(n) = $3 \cdot 2^{n-1}$